

Logistic Regression

From linear scores → probabilities → Binary Cross-Entropy

The main idea:

We want a model that predicts a probability between 0 and 1 for a binary outcome.

Example: $P(y = 1 \mid x) = 0.82$

Start with a Linear Combination

$$z = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

This is exactly like a linear model.

Problem: z can be any value ($-\infty$ to $+\infty$).

But a probability must lie between 0 and 1.

So what should the linear score represent?

$$\log(p / (1 - p)) = z$$

Instead of making z the probability, we make z the LOG-ODDS.

$$\text{Odds} = p / (1 - p)$$

Taking log lets the quantity range from $-\infty$ to $+\infty$, just like z .

Convert log-odds back to probability

$$p = 1 / (1 + e^{-z})$$

This is the sigmoid function.

It takes any real-valued z and maps it to $(0, 1)$.

Therefore: linear model $\rightarrow$ sigmoid $\rightarrow$ probability

Now: How do we train the model?

For one example, the model predicts $p = P(y = 1 \mid x)$.

$$P(y \mid x) = p^y (1 - p)^{1-y}$$

If $y = 1 \rightarrow \text{likelihood} = p$

If $y = 0 \rightarrow \text{likelihood} = 1 - p$

So the likelihood gives high value when the model assigns high probability to the correct class.

Maximum Likelihood → Log-Likelihood

For N independent training examples, multiply their likelihoods:

$$L = \prod_i p_i^{y_i} (1 - p_i)^{1-y_i}$$

Products are inconvenient, so take log. Maximizing $\log(L)$ gives the same optimum.

$$\log L = \sum_i [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

And this is exactly where BCE comes from

Training usually MINIMIZES a loss, so we take the negative log-likelihood:

$$\text{Loss} = - \sum_i [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

Divide by N → Binary Cross-Entropy (BCE)

$$\text{BCE} = -(1/N) \sum_i [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

The whole story in one chain

Features → Linear score → Sigmoid → Probability → BCE

$$z = \mathbf{w}^T \mathbf{x} + b$$

$$p = \sigma(z) = 1/(1+e^{-z})$$

Choose w and b that minimize BCE $\Leftrightarrow$ maximize likelihood

One simple intuition

Suppose the true label is $y = 1$.

Prediction $p = 0.95$

→ low BCE loss

Prediction $p = 0.05$

→ very high BCE loss

BCE strongly penalizes being confidently wrong.

Logistic Regression — the takeaway

1. Build a linear score

$$z = \mathbf{w}^T \mathbf{x} + b$$

2. Interpret it as log-odds and convert it to probability with sigmoid

3. Maximum likelihood gives the log-likelihood

4. Negating it gives the loss we minimize

→ **Binary Cross-Entropy**